



Chapter 20

Network Layer: Internet Protocol

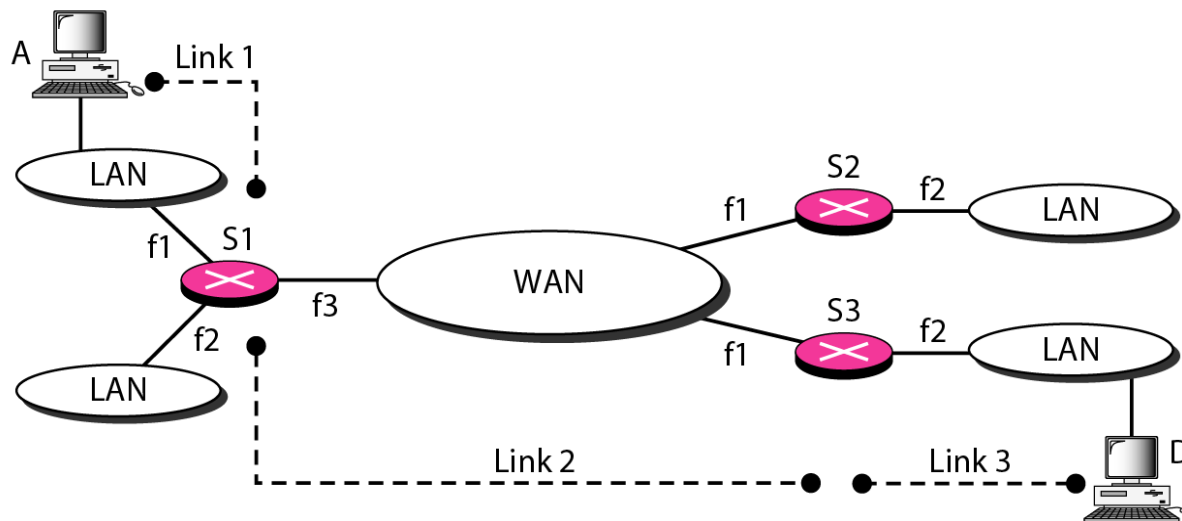
Introduction

- Main network protocol is **Internet Protocol(IP)**.
- Current version of IP is 4 or IPv4.
- Next Generation of this protocol is **IPv6** which may become the dominant protocol in the near future.

20-1 INTERNETWORKING

- Physical and Data link layer of network operate locally.
- These two layers are responsible for data delivery on network from one node to next.

Figure 20.1 *Links between two hosts*

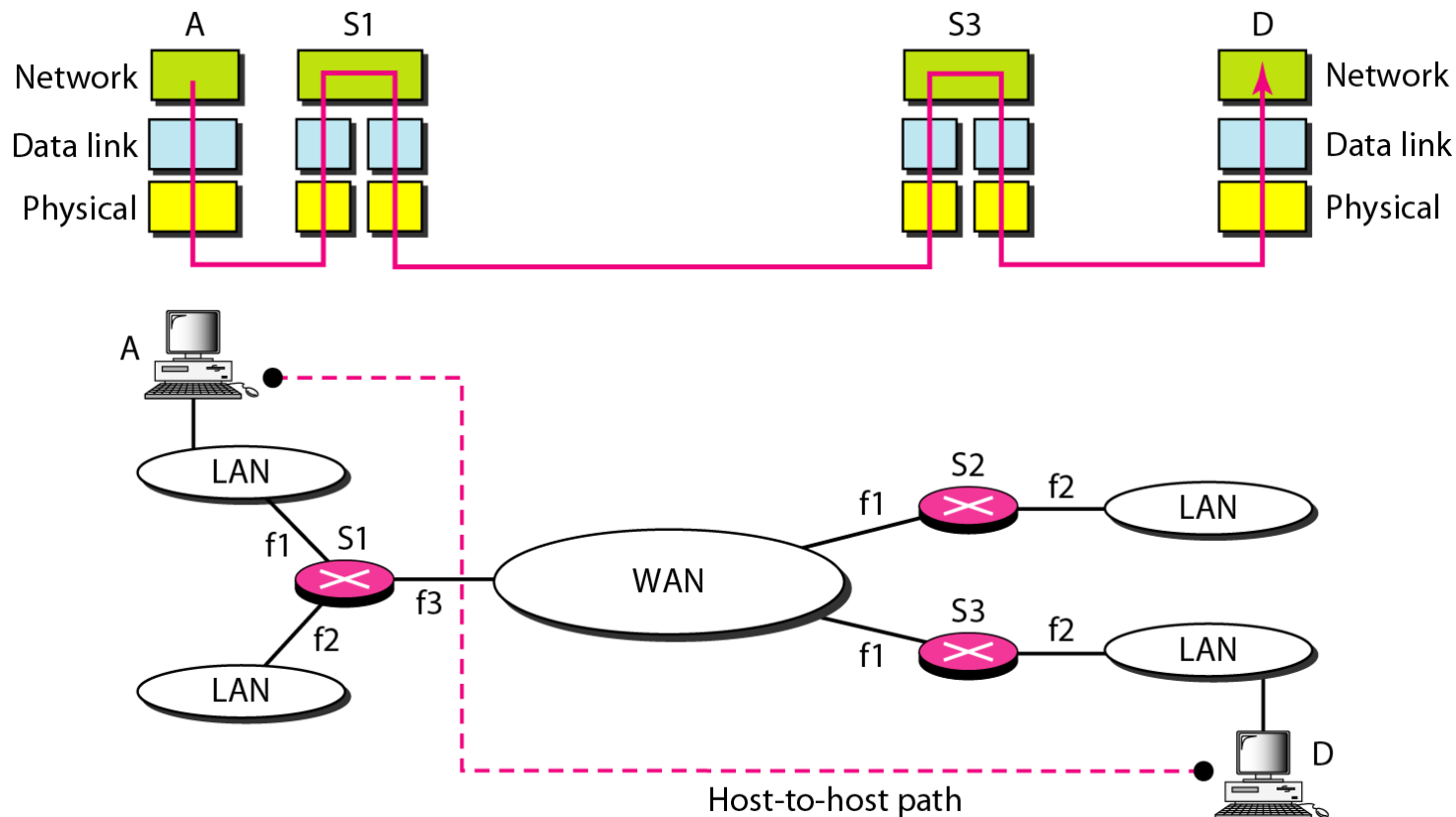


- Internetwork is made of five networks: Four LANs and one WAN.
- If host A need to send a data packet to host D, the packet needs to go first from A to R1, then from R1 to R3 and finally from R3 to D.
- **Problem:** When data arrives at interface f1 of R1, how does R1 know that interface f3 is the outgoing interface?
- There is no provision in data link layer or physical layer to help R1 to make right decision. Also frame does no contain any routing information.

Need of Network Layer

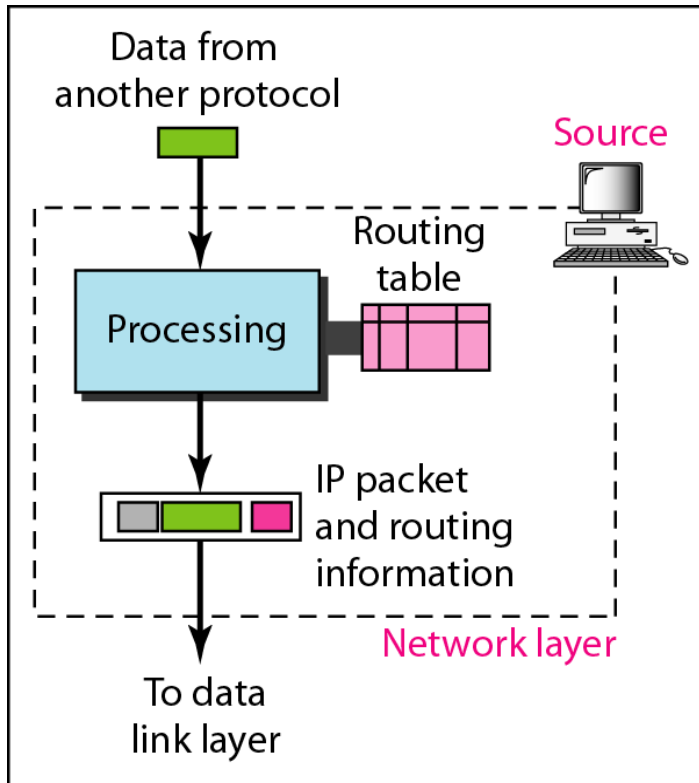
- To solve the problem of delivery through several links, the network layer was designed .
- The network layer is responsible for host to host delivery and for routing the packets through routers or switches.

Figure 20.2 *Network layer in an internetwork*

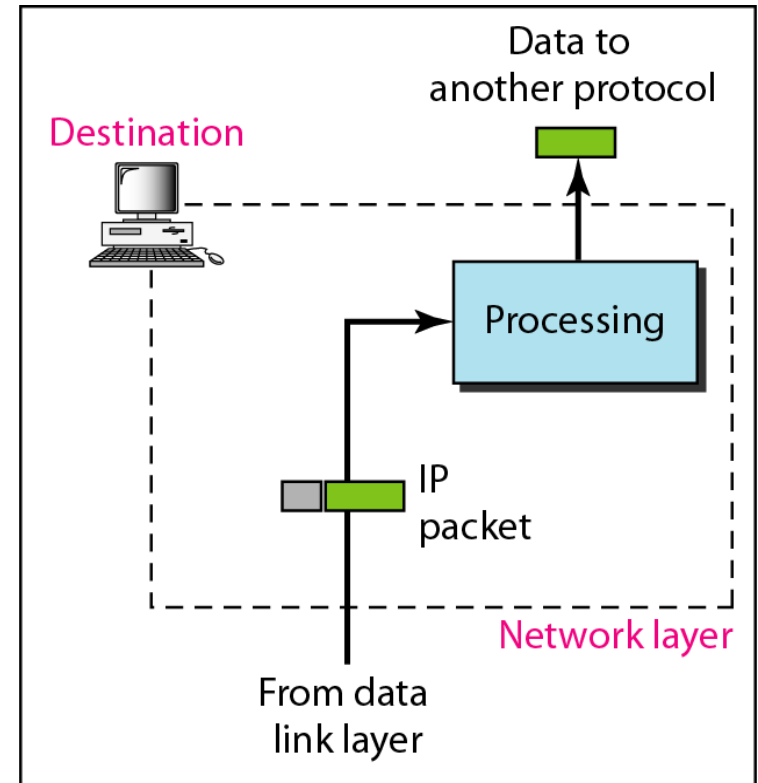


- The network layer is responsible for creating the packet from the data coming from transport layer.
- The header contains the logical address of source and destination.
- The network layer is responsible for checking its routing table to find the routing information(outgoing interface of packet etc.).
- The network layer at the switch or router is responsible for routing the packet.
- When packet arrives the router or switch consults its routing table and finds the interface from which the packet must be sent.

Figure 20.3 *Network layer at the source, router, and destination*

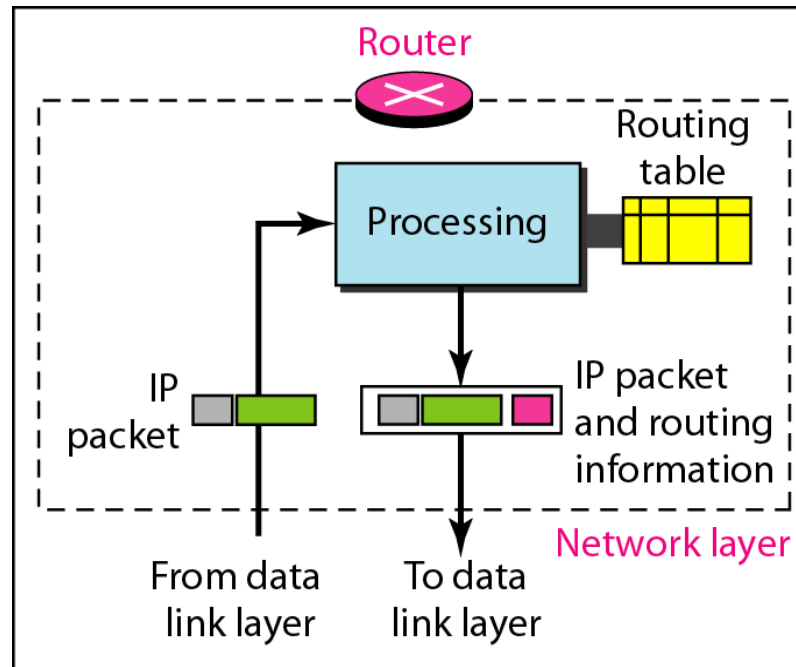


a. Network layer at source



b. Network layer at destination

Figure 20.3 *Network layer at the source, router, and destination (continued)*



c. Network layer at a router

Internet as a Datagram Network

- The internet at the network layer is a packet switched network.
- Switching: Circuit, Packet & Message.
- Packet Switching uses either Virtual circuit or datagram approach.

Switching at the network layer in the Internet uses the datagram approach to packet switching.

Internet as a Connectionless Network

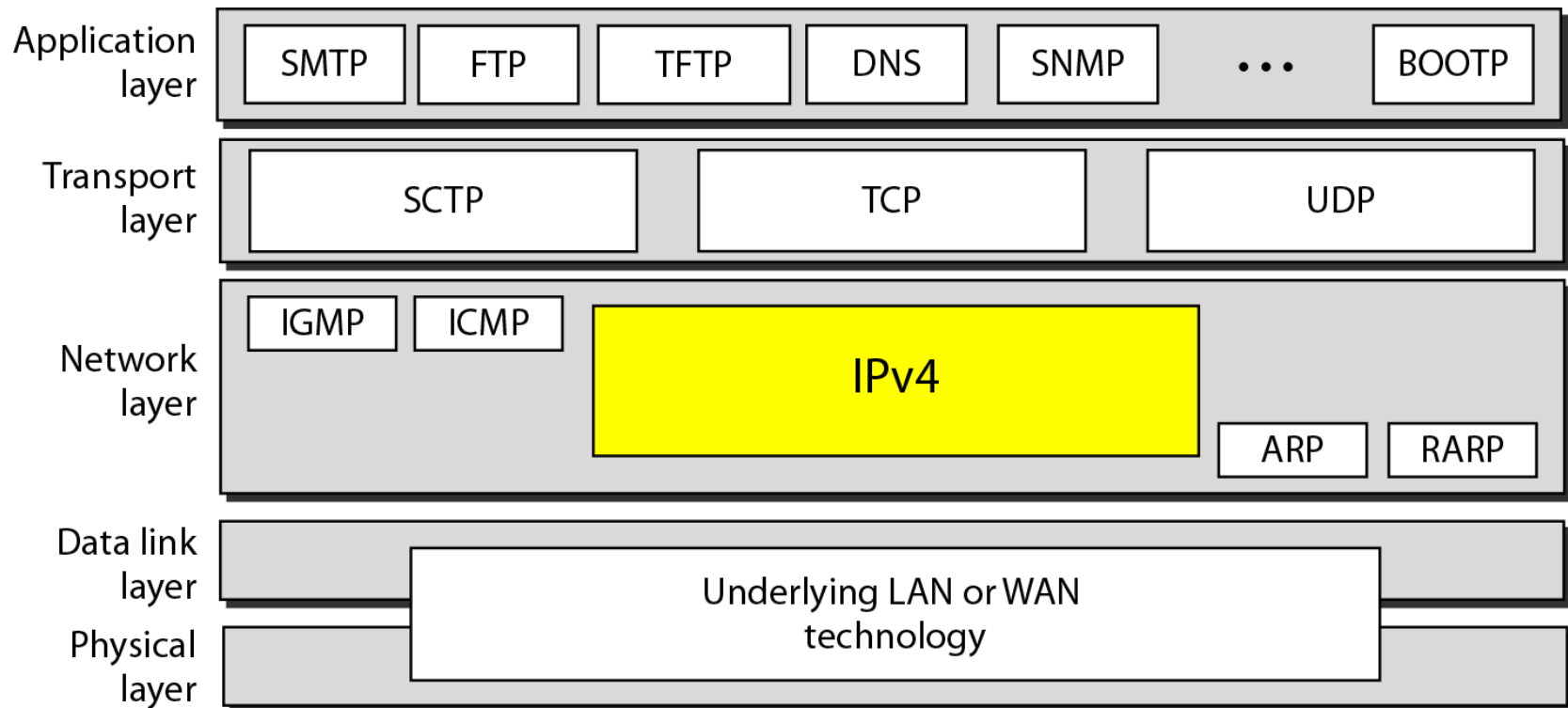
- Delivery of packet can be accomplished by using either connection-oriented or a connectionless network service.
- In connection-oriented service a connection is setup before sending the packet. Packets are sent on the same path in sequential order.
- In connectionless service, the network layer protocol treats each packet independently.

Switching at the network layer in the Internet uses the datagram approach to packet switching.

20-2 IPv4

*The Internet Protocol version 4 (**IPv4**) is the delivery mechanism used by the TCP/IP protocols.*

Figure 20.4 *Position of IPv4 in TCP/IP protocol suite*

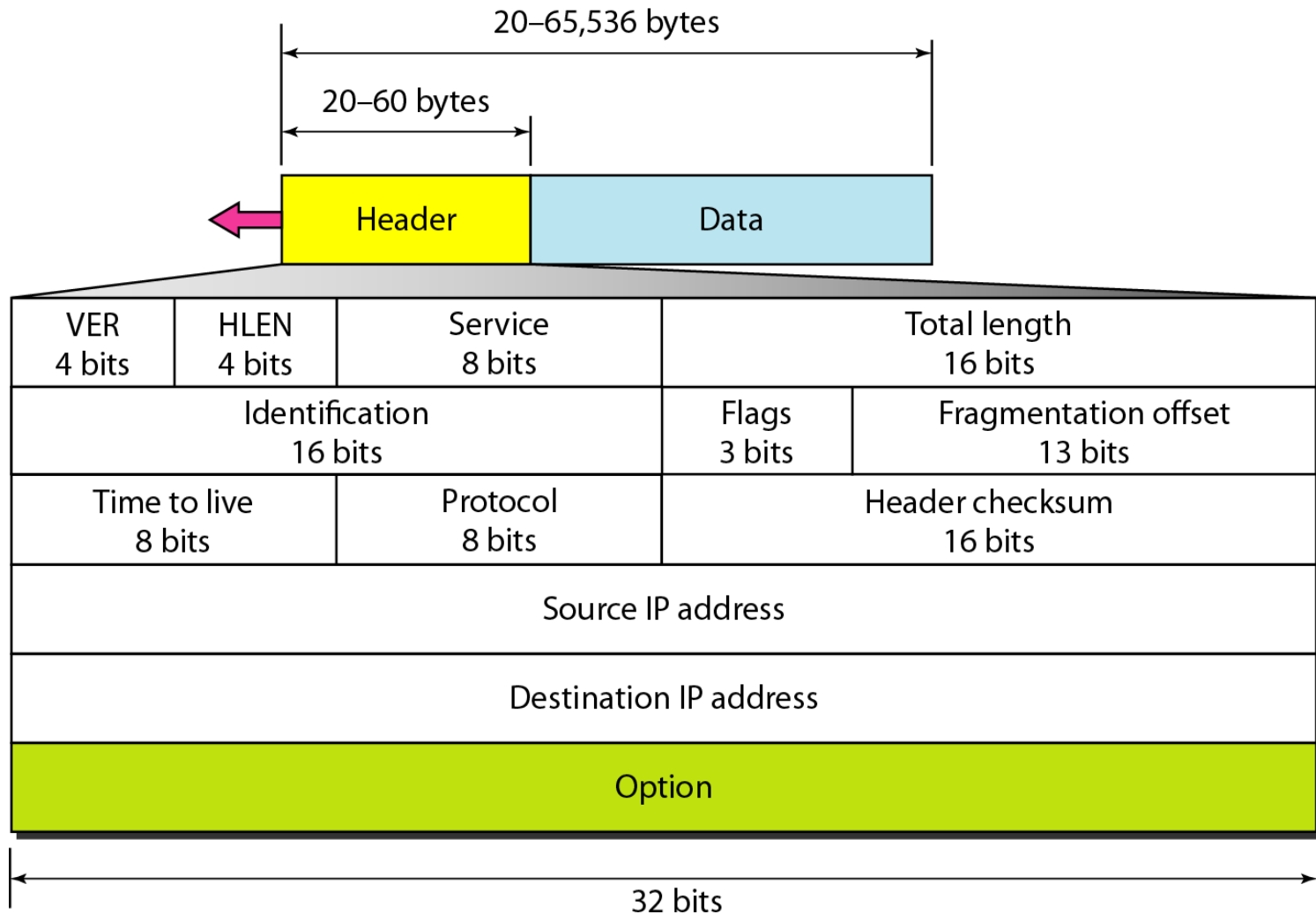


- IPv4 is unreliable and connectionless datagram protocol- a best delivery service.
- Best effort means IPv4 provides no error and flow control.
- For reliability IPv4 is paired with a reliable protocol such as TCP.
- IPv4 uses connectionless protocol for a packet switching network that uses datagram approach.
- IPv4 relies on higher level protocol to take care of all these problems.

Datagram

Packets in the IPv4 layer are called **datagrams**.

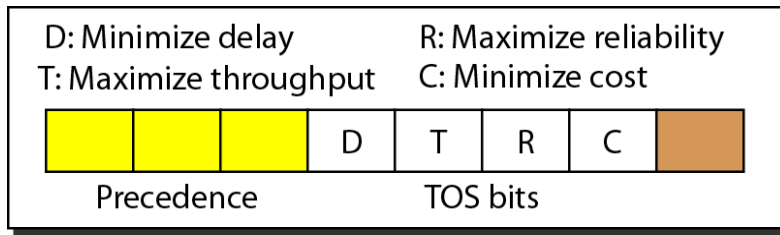
Figure 20.5 *IPv4 datagram format*



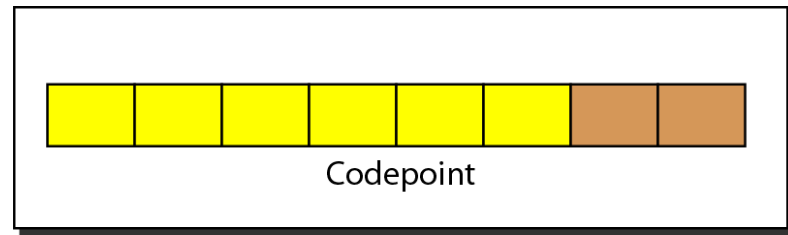
Datagram continues.....

- Datagram is a variable length packet consisting of two parts:
header and data.
- Header is 20 to 60 bytes in length and contains routing and delivery information.
- Description of each field:
 1. **Version(VER):** 4-bit field defines the version of IPv4 protocol. Current version is 4.
 2. **Header Length(HLEN):** 4-bit field defines the total length of the datagram header in 4-byte words.
If there are no options the header length is 20 bytes and value of this field is 5 ($5 \times 4 = 20$).
When option field is at its maximum size, the value of this field is 15 ($15 \times 4 = 60$).
 3. **Services:** This field previously called service type and now it is called differential services.

Figure 20.6 *Service type or differentiated services*



Service type



Differentiated services

Service type:

The first 3 bits are called precedence bits and next 4 bits are called type of service and last bit is not used.

- Precedence** is a 3-bit subfield ranging from 000 to 111. The precedence defines the priority of the datagram in issues such as congestion. If router is congested and needs to discard some datagrams, those datagrams with lowest precedence are discarded first. The precedence subfield was part of version 4, but never used.
- TOS bits** is a 4-bit subfield with each bit having a special meaning.

Table 20.1 *Types of service*

<i>TOS Bits</i>	<i>Description</i>
0000	Normal (default)
0001	Minimize cost
0010	Maximize reliability
0100	Maximize throughput
1000	Minimize delay

Differential services:

First 6 bits make up the codepoint subfield and the last 2 bits are not used.

- a) When 3 rightmost bits are 0, the 3 leftmost bits are interpreted the same as the precedence bits.
- b) When 3 rightmost bits are not all 0, the 6 bits define 64 services based on priority assignment by the internet or local authorities.

Table 20.3 *Values for codepoints*

<i>Category</i>	<i>Codepoint</i>	<i>Assigning Authority</i>
1	XXXXX0	Internet
2	XXXXX11	Local
3	XXXXX01	Temporary or experimental

4. Total Length: 16-bit field that defines the total length(header plus data) of the IPv4 datagrams in bytes.

Length of data = Total length - header length

Since the field length is 16 bits, the total length of the IPv4 datagram is limited to 65,536 bytes of which 20 to 60 bytes are header and rest is data from upper layer.

5. Identification: This field is used in fragmentation.

6. Flags: This field is used in fragmentation.

7. Fragmentation offset: This field is used in fragmentation.

Fragmentation

Maximum Transfer Unit (MTU)

Each data link layer protocol has its own frame format in most protocols. One field is the maximum size of the data field. When datagram is encapsulated in a frame, the total size of the datagram must be less than this maximum size, which is defined by the hardware and software used in the network.

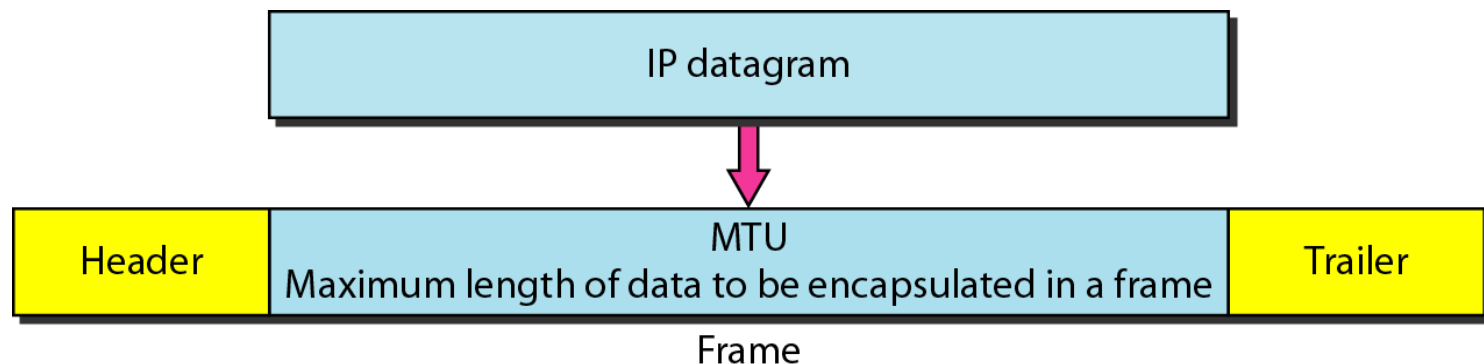


Table 20.5 *MTUs for some networks*

<i>Protocol</i>	<i>MTU</i>
Hyperchannel	65,535
Token Ring (16 Mbps)	17,914
Token Ring (4 Mbps)	4,464
FDDI	4,352
Ethernet	1,500
X.25	576
PPP	296

- To make the IPv4 protocol independent of the physical network, the designers decided to make the maximum length of the IPv4 datagram equal to 65,535 bytes.
- For other physical networks, we must divide the datagram to make it possible to pass through these networks. This is called **fragmentation**.
- When a datagram is fragmented, each fragment has its own header with most of the fields repeated, but with some changed. A fragmented datagram may itself be fragmented if it encounters a network with an even smaller MTU.
- The reassembly of the datagram, is done only by the destination host because each fragment becomes an independent datagram.

Fields Related to Fragmentation

5. Identification:

- This 16-bit field identifies a datagram originating from the source host. The combination of the identification and source IPv4 address must uniquely define a datagram as it leaves the source host.
- When a datagram is fragmented, the value in the identification field is copied to all fragments. All fragments have the same identification number, the same as the original datagram.
- The identification number helps the destination in reassembling the datagram.

6. Flags:

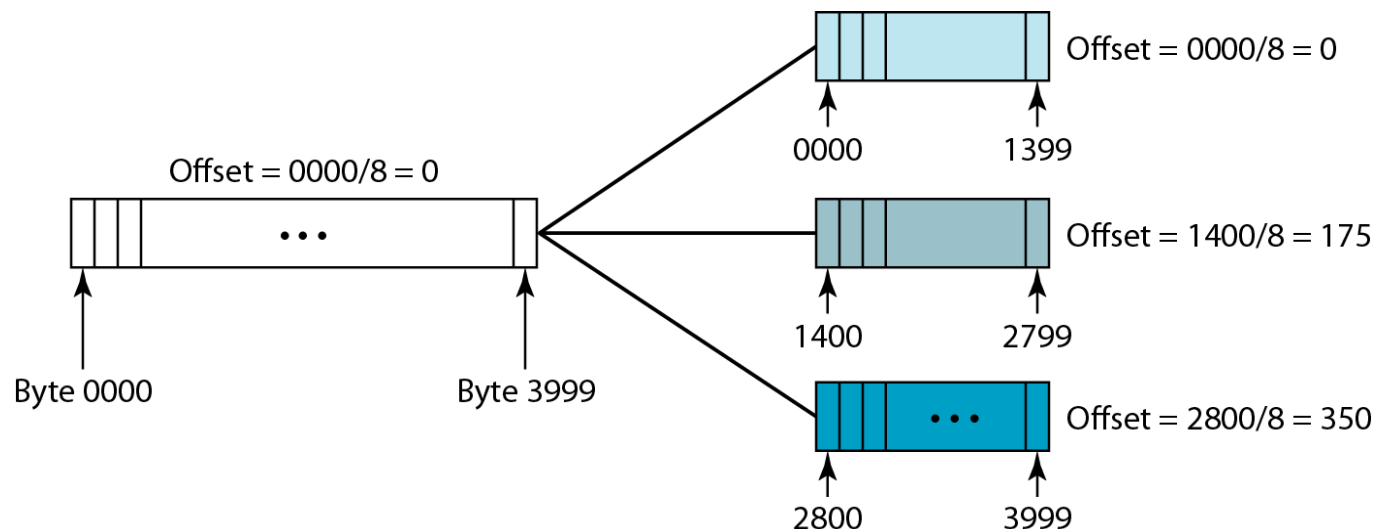
- This is a 3-bit field. The first bit is reserved.
- The second bit is called the do not fragment bit.
 1. If its value is 1, the machine must not fragment the datagram. If it cannot pass the datagram through any available physical network, it discards the datagram and sends an ICMP error message to the source host.
 2. If its value is 0, the datagram can be fragmented if necessary.
- The third bit is called the more fragment bit.
 1. If its value is 1, it means the datagram is not the last fragment; there are more fragments after this one.
 2. If its value is 0, it means this is the last or only fragment

Figure 20.10 *Flags used in fragmentation*



7. Fragmentation Offset:

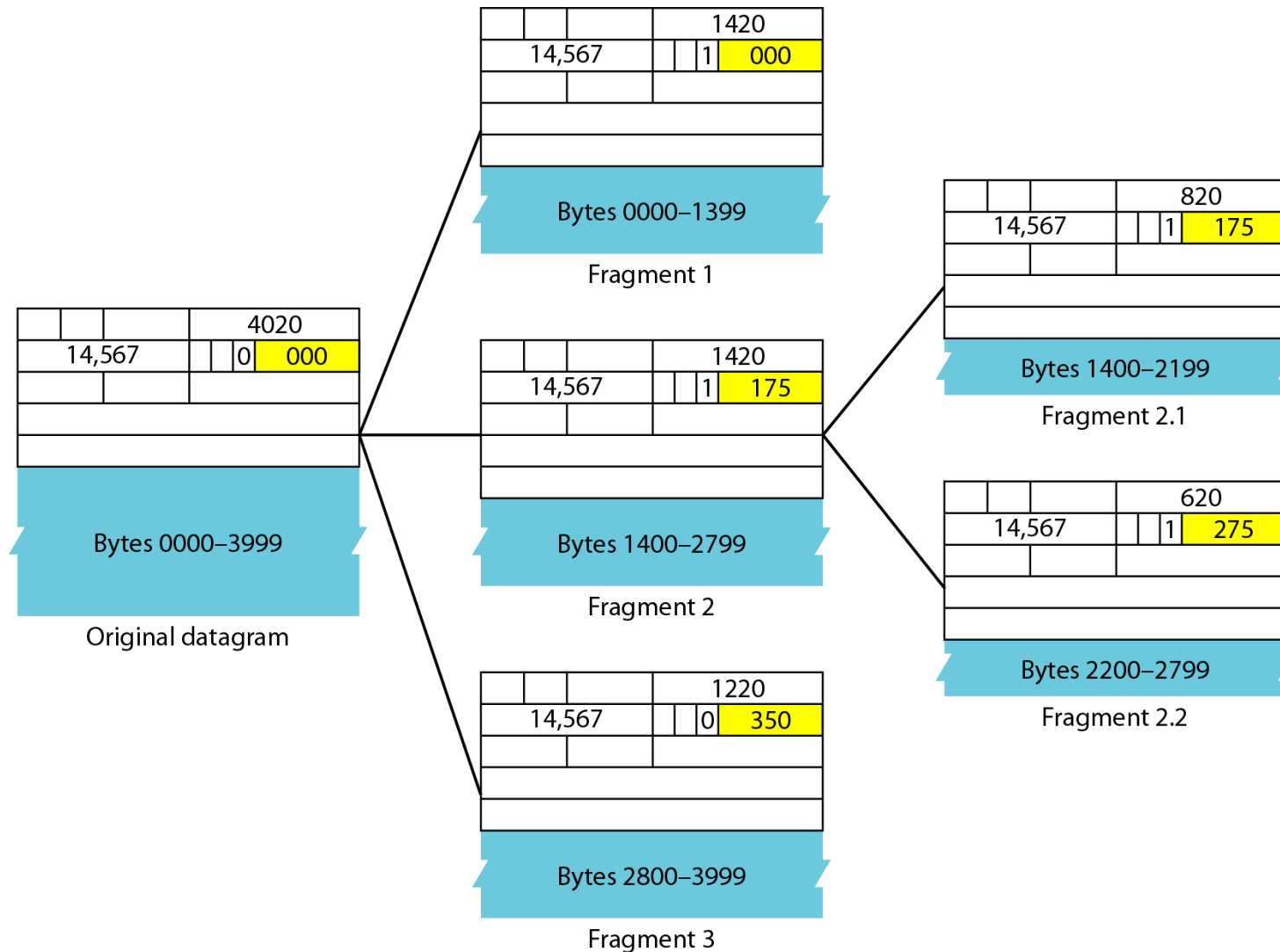
- 13-bit field shows relative position of fragment with respect to whole datagram. It is the offset of the data in the original datagram measured in units of 8 bytes.
- Figure 20.11 shows a datagram with a data size of 4000 bytes fragmented into three fragments.



- The bytes in the original datagram are numbered 0 to 3999.
- The first fragment carries bytes 0 to 1399. The offset is $0/8 = 0$.
- The second fragment carries bytes 1400 to 2799; the offset is $1400/8 = 175$.
- Finally, the third fragment carries bytes 2800 to 3999. The offset is $2800/8 = 350$.
- Remember that the value of the offset is measured in units of 8 bytes. This is done because the length of the offset field is only 13 bits and cannot represent a sequence of bytes greater than 8191. This forces hosts or routers that fragment datagrams to choose a fragment size so that the first byte number is divisible by 8.

- Notice the value of the identification field is the same in all fragments. Notice the Value of the flags field with the more bit set for all fragments except the last. Also, the value of the offset field for each fragment is shown.

Figure 20.12 *fragmentation example*



7. Time to live:

- A datagram has a limited lifetime in its travel through an internet.

This field was originally designed to hold a timestamp, which was decremented by each visited router. The datagram was discarded when the value became zero.

- This field is needed because routing tables in the Internet can become corrupted. A datagram may travel between two or more routers for a long time without ever getting delivered to the destination host. This field limits the lifetime of a datagram.

8. Protocol:

- This 8-bit field defines the higher-level protocol that uses the services of the IPv4 layer.
- An IPv4 datagram can encapsulate data from several higher-level protocols such as TCP, UDP, ICMP, and IGMP.
- This field specifies the final destination protocol to which the IPv4 datagram is delivered.

Figure 20.8 *Protocol field and encapsulated data*

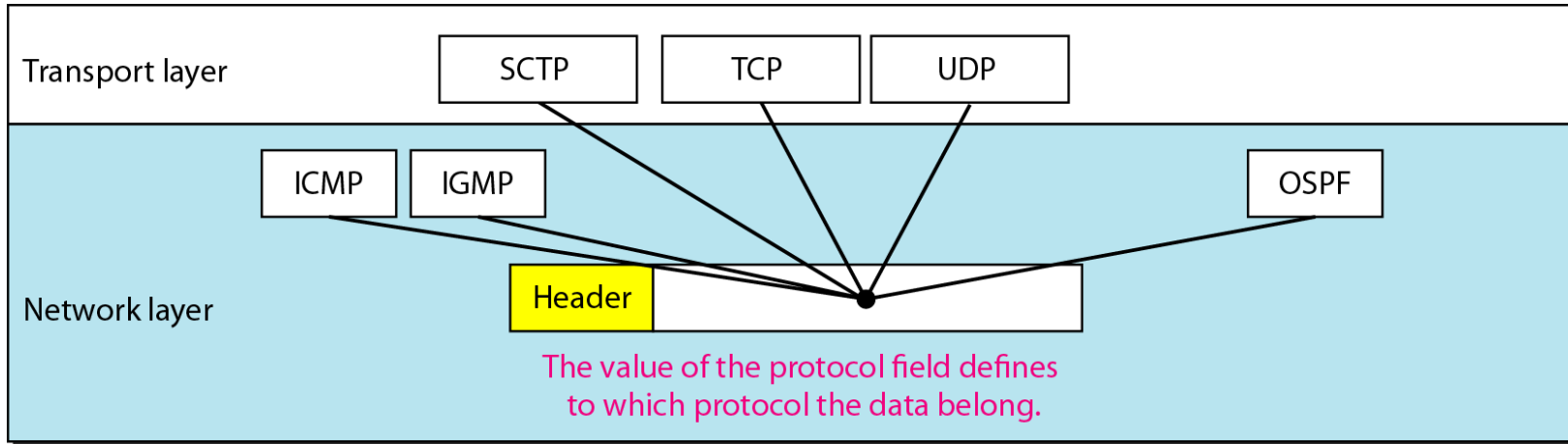


Table 20.4 *Protocol values*

<i>Value</i>	<i>Protocol</i>
1	ICMP
2	IGMP
6	TCP
17	UDP
89	OSPF

9. Checksum:

- First, the value of the checksum field is set to 0. Then the entire header is divided into 16-bit sections and added together. The result (sum) is complemented and inserted into the checksum field.

10. Source address:

- This 32-bit field defines the IPv4 address of the source. This field must remain unchanged during the time the IPv4 datagram travels from the source host to the destination host.

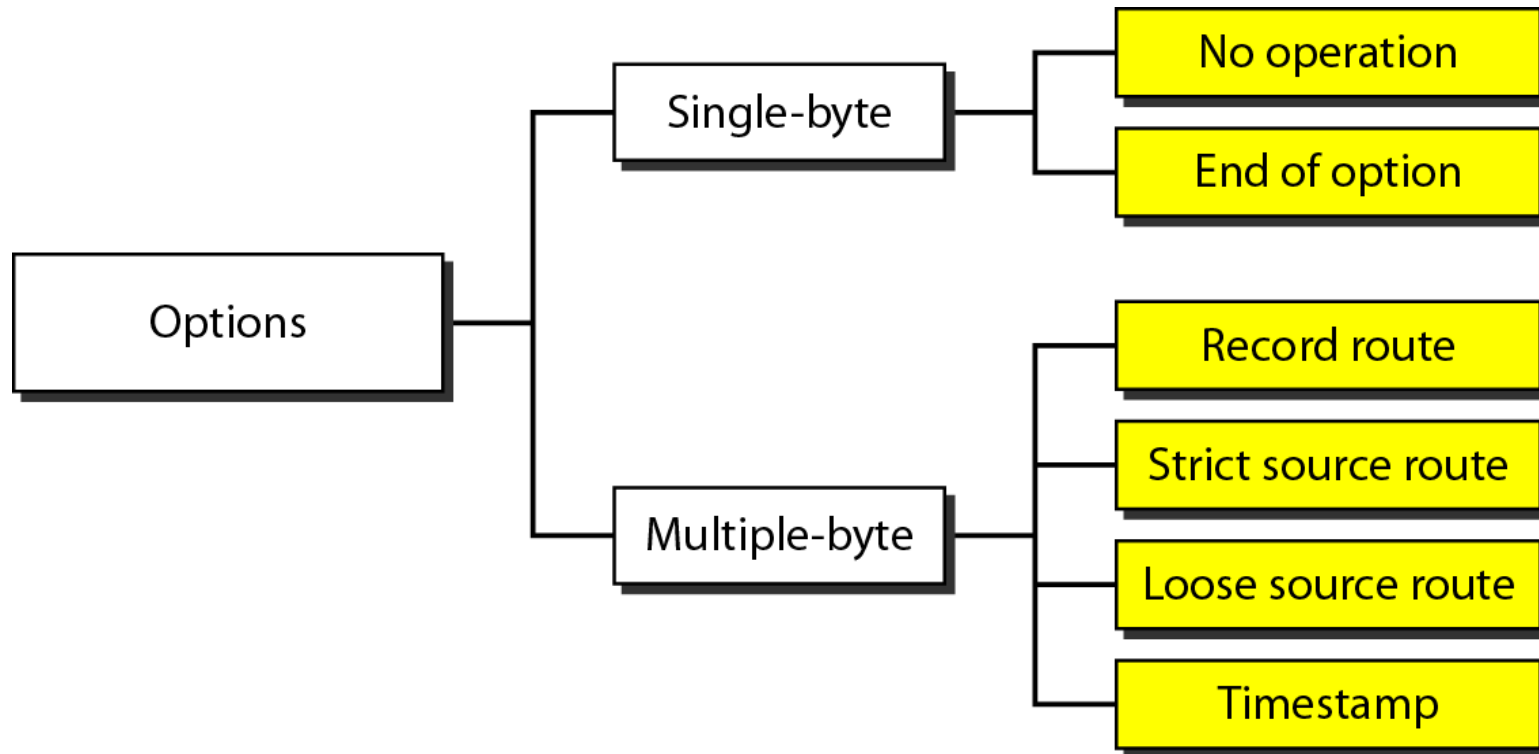
11. Destination address:

- This 32-bit field defines the IPv4 address of the destination. This field must remain unchanged during the time the IPv4 datagram travels from the source host to the destination host.

12. Options:

- The header of the IPv4 datagram is made of two parts: a fixed part and a variable part. The fixed part is 20 bytes long and the variable part comprises the options that can be a maximum of 40 bytes.
- Options, as the name implies, are not required for a datagram. They can be used for network testing and debugging.

Figure 20.14 *Taxonomy of options in IPv4*





Example 20.1

An IPv4 packet has arrived with the first 8 bits as shown:

01000010

The receiver discards the packet. Why?

Solution

There is an error in this packet. The 4 leftmost bits (0100) show the version, which is correct. The next 4 bits (0010) show an invalid header length ($2 \times 4 = 8$). The minimum number of bytes in the header must be 20. The packet has been corrupted in transmission.



Example 20.2

In an IPv4 packet, the value of HLEN is 1000 in binary. How many bytes of options are being carried by this packet?

Solution

The HLEN value is 8, which means the total number of bytes in the header is 8×4 , or 32 bytes. The first 20 bytes are the base header, the next 12 bytes are the options.



Example 20.3

In an IPv4 packet, the value of HLEN is 5, and the value of the total length field is 0x0028. How many bytes of data are being carried by this packet?

Solution

The HLEN value is 5, which means the total number of bytes in the header is 5×4 , or 20 bytes (no options). The total length is 40 bytes, which means the packet is carrying 20 bytes of data ($40 - 20$).



Example 20.4

An IPv4 packet has arrived with the first few hexadecimal digits as shown.

0x45000028000100000102...

How many hops can this packet travel before being dropped? The data belong to what upper-layer protocol?

Solution

To find the time-to-live field, we skip 8 bytes. The time-to-live field is the ninth byte, which is 01. This means the packet can travel only one hop. The protocol field is the next byte (02), which means that the upper-layer protocol is IGMP.



Example 20.5

A packet has arrived with an M bit value of 0. Is this the first fragment, the last fragment, or a middle fragment? Do we know if the packet was fragmented?

Solution

If the M bit is 0, it means that there are no more fragments; the fragment is the last one. However, we cannot say if the original packet was fragmented or not. A non-fragmented packet is considered the last fragment.



Example 20.6

A packet has arrived with an M bit value of 1. Is this the first fragment, the last fragment, or a middle fragment? Do we know if the packet was fragmented?

Solution

If the M bit is 1, it means that there is at least one more fragment. This fragment can be the first one or a middle one, but not the last one. We don't know if it is the first one or a middle one; we need more information (the value of the fragmentation offset).



Example 20.7

A packet has arrived with an M bit value of 1 and a fragmentation offset value of 0. Is this the first fragment, the last fragment, or a middle fragment?

Solution

Because the M bit is 1, it is either the first fragment or a middle one. Because the offset value is 0, it is the first fragment.



Example 20.8

A packet has arrived in which the offset value is 100. What is the number of the first byte? Do we know the number of the last byte?

Solution

To find the number of the first byte, we multiply the offset value by 8. This means that the first byte number is 800. We cannot determine the number of the last byte unless we know the length.



Example 20.9

A packet has arrived in which the offset value is 100, the value of HLEN is 5, and the value of the total length field is 100. What are the numbers of the first byte and the last byte?

Solution

The first byte number is $100 \times 8 = 800$. The total length is 100 bytes, and the header length is 20 bytes (5×4), which means that there are 80 bytes in this datagram. If the first byte number is 800, the last byte number must be 879.

20-3 IPv6

- The network layer protocol in the TCP/IP protocol suite is currently IPv4. Although IPv4 is well designed, data communication has evolved since the inception of IPv4 in the 1970s.
- IPv4 has some **deficiencies** that make it unsuitable for the fast-growing Internet.
 1. Address depletion is a long-term problem in the Internet.
 2. The Internet must accommodate real-time audio and video transmission. This type of transmission requires minimum delay strategies and reservation of resources not provided in the IPv4 design.
 3. The Internet must accommodate encryption and authentication of data for some applications. No encryption or authentication is provided by IPv4.
- To overcome these deficiencies, IPv6 (Internetworking Protocol, version 6), also known as IPng (Internetworking Protocol, next generation), was proposed and is now a standard.

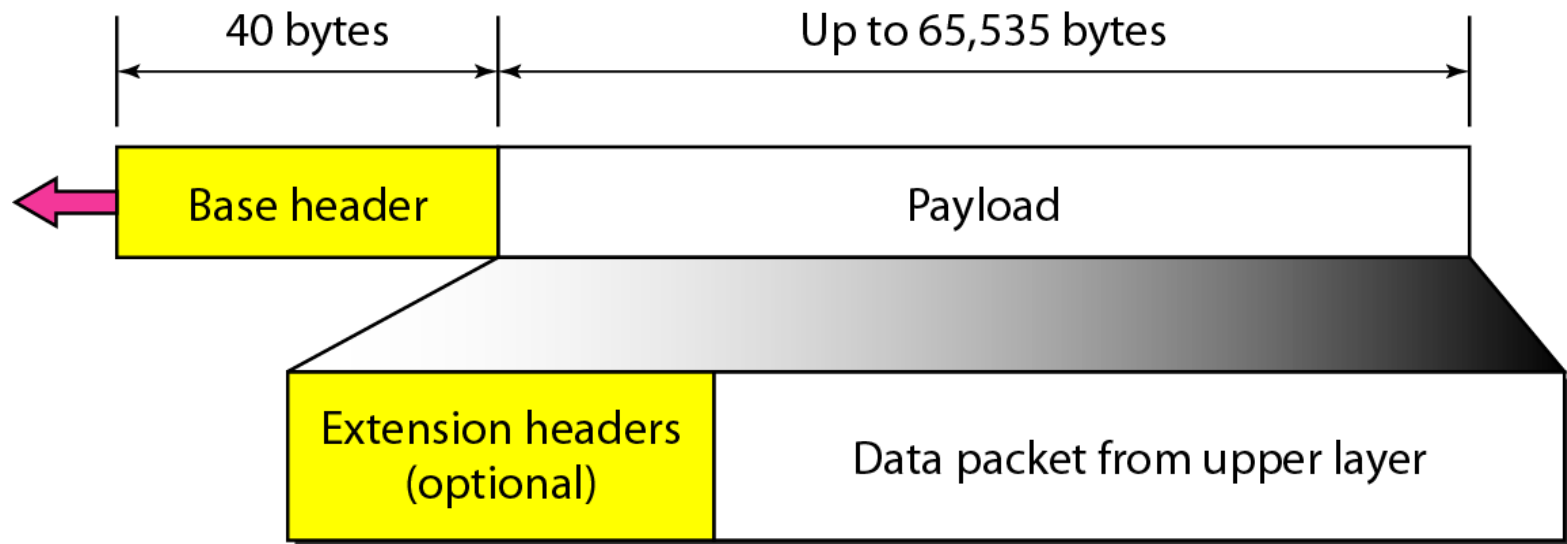
Advantages:

- Larger address space. An IPv6 address is 128 bits long. Compared with the 32-bit address of IPv4, this is a huge the address space.
- Better header format.
- New options. IPv6 has new options to allow for additional functionalities.
- Allowance for extension. IPv6 is designed to allow the extension of the protocol if required by new technologies or applications.
- Support for resource allocation.
- Support for more security.

Packet Format

- Each packet is composed of a mandatory base header followed by the payload.
- The payload consists of two parts: optional extension headers and data from an upper layer.
- The base header occupies 40 bytes, whereas the extension headers and data from the upper layer contain up to 65,535 bytes of information.

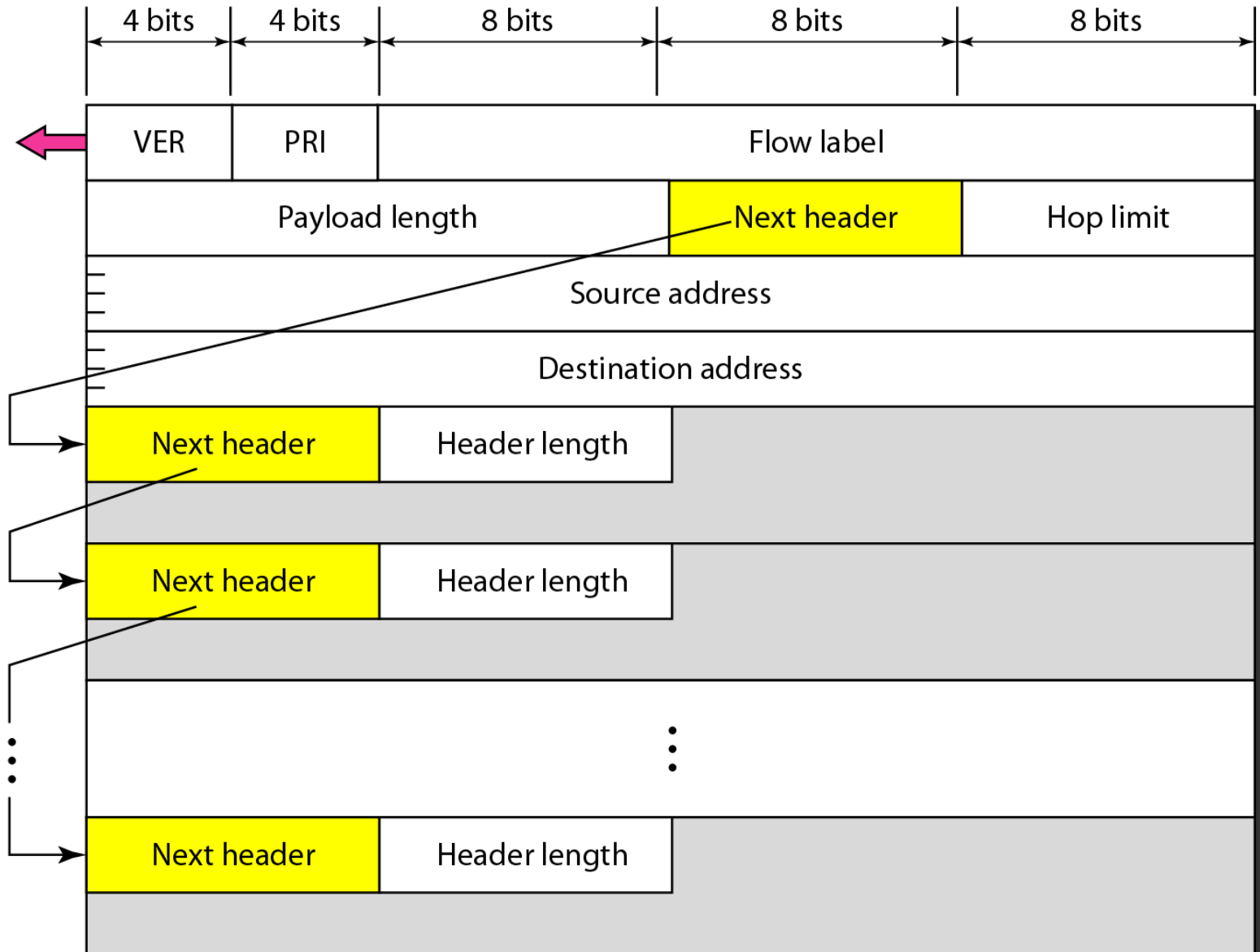
Figure 20.15 *IPv6 datagram header and payload*



Base Header

- There are 8 fields in Base Header

Figure 20.16 *Format of an IPv6 datagram*



- Version.** This 4-bit field defines the version number of the IP. For IPv6, the value is 6.
- Priority.** The 4-bit priority field defines the priority of the packet with respect to traffic congestion.
- Flow label.** The flow label is a 3-byte (24-bit) field that is designed to provide special handling for a particular flow of data.

A sequence of packets, sent from a particular source to a particular destination, that needs special handling by routers is called a **flow of packets**. The combination of the source address and the value of the flow label uniquely defines a flow of packets.

- Payload length.** The 2-byte payload length field defines the length of the IP datagram excluding the base header.

•**Next header.** The next header is an 8-bit field defining the header that follows the base header in the datagram.

The next header is either one of the optional extension headers used by IP or the header of an encapsulated packet such as UDP or TCP.

Note: This field in version 4 is called the protocol.

Table 20.6 *Next header codes for IPv6*

<i>Code</i>	<i>Next Header</i>
0	Hop-by-hop option
2	ICMP
6	TCP
17	UDP
43	Source routing
44	Fragmentation
50	Encrypted security payload
51	Authentication
59	Null (no next header)
60	Destination option

- Hop limit.** This 8-bit hop limit field serves the same purpose as the TTL field in IPv4.
- Source address.** The source address field is 16-byte (128-bit) Internet address that identifies original source of the datagram.
- Destination address.** The destination address field is a 16-byte (128-bit) Internet address that usually identifies the final destination of the datagram.

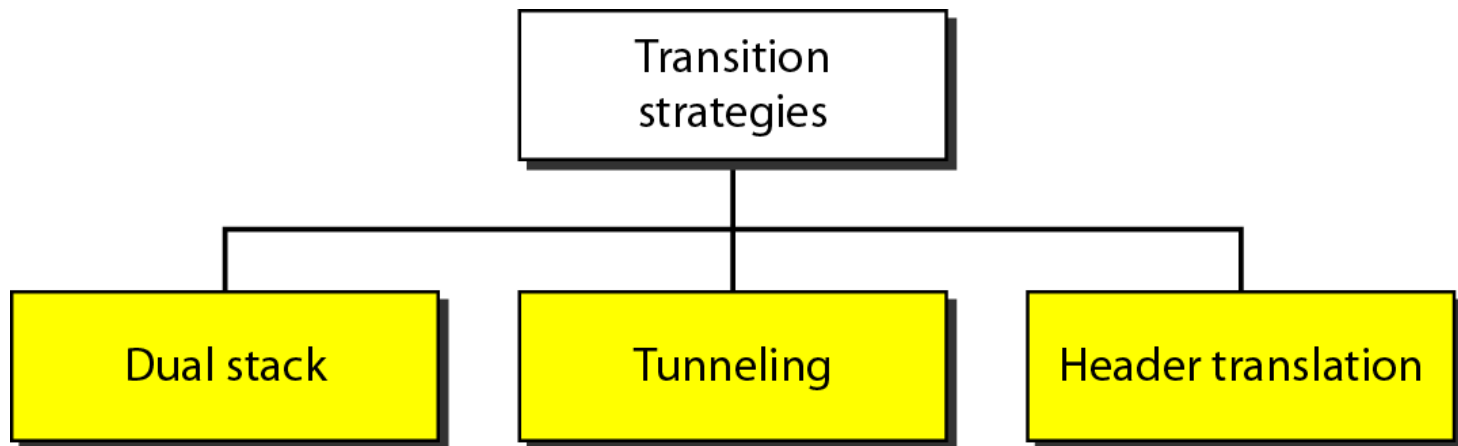
Table 20.9 *Comparison between IPv4 and IPv6 packet headers*

<i>Comparison</i>
1. The header length field is eliminated in IPv6 because the length of the header is fixed in this version.
2. The service type field is eliminated in IPv6. The priority and flow label fields together take over the function of the service type field.
3. The total length field is eliminated in IPv6 and replaced by the payload length field.
4. The identification, flag, and offset fields are eliminated from the base header in IPv6. They are included in the fragmentation extension header.
5. The TTL field is called hop limit in IPv6.
6. The protocol field is replaced by the next header field.
7. The header checksum is eliminated because the checksum is provided by upper-layer protocols; it is therefore not needed at this level.
8. The option fields in IPv4 are implemented as extension headers in IPv6.

20-4 TRANSITION FROM IPv4 TO IPv6

- Because of the huge number of systems on the Internet, the transition from IPv4 to IPv6 cannot happen suddenly. It takes a considerable amount of time before every system in the Internet can move from IPv4 to IPv6.
- The transition must be smooth to prevent any problems between IPv4 and IPv6 systems.
- Three strategies have been devised by the IETF to help the transition

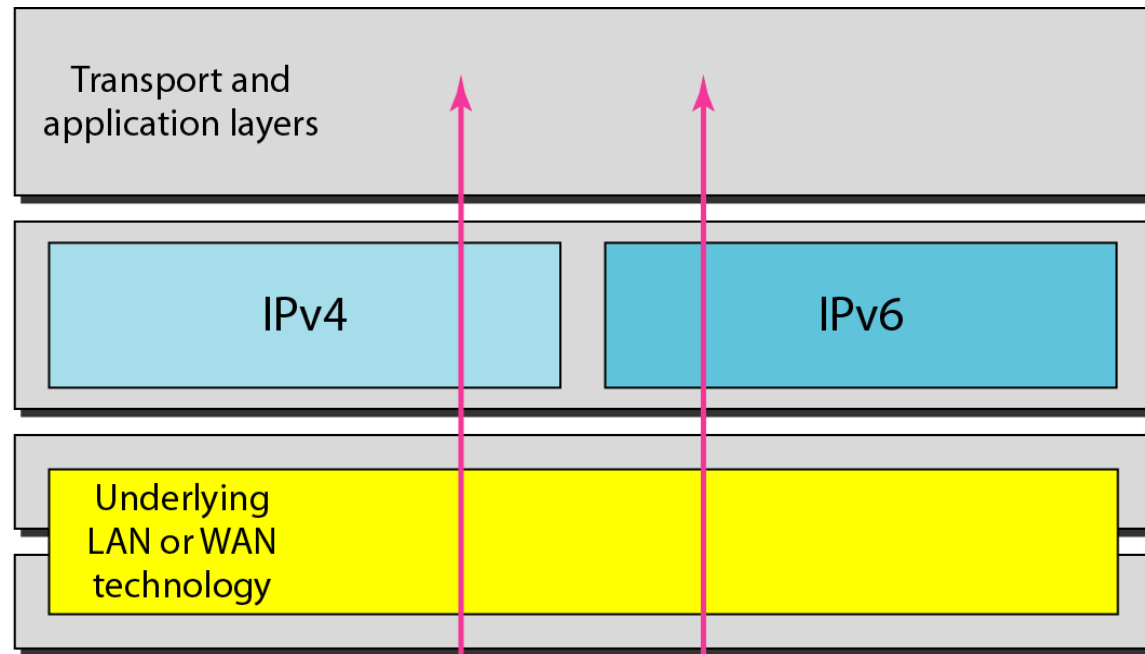
Figure 20.18 *Three transition strategies*



Dual Stack

- It is recommended that all hosts, have a dual stack of protocols. A station must run IPv4 and IPv6 simultaneously until all the Internet uses IPv6.
- To determine which version to use when sending a packet to a destination, the source host queries the DNS. If the DNS returns an IPv4 address, the source host sends an IPv4 packet. If the DNS returns an IPv6 address, the source host sends an IPv6 packet.

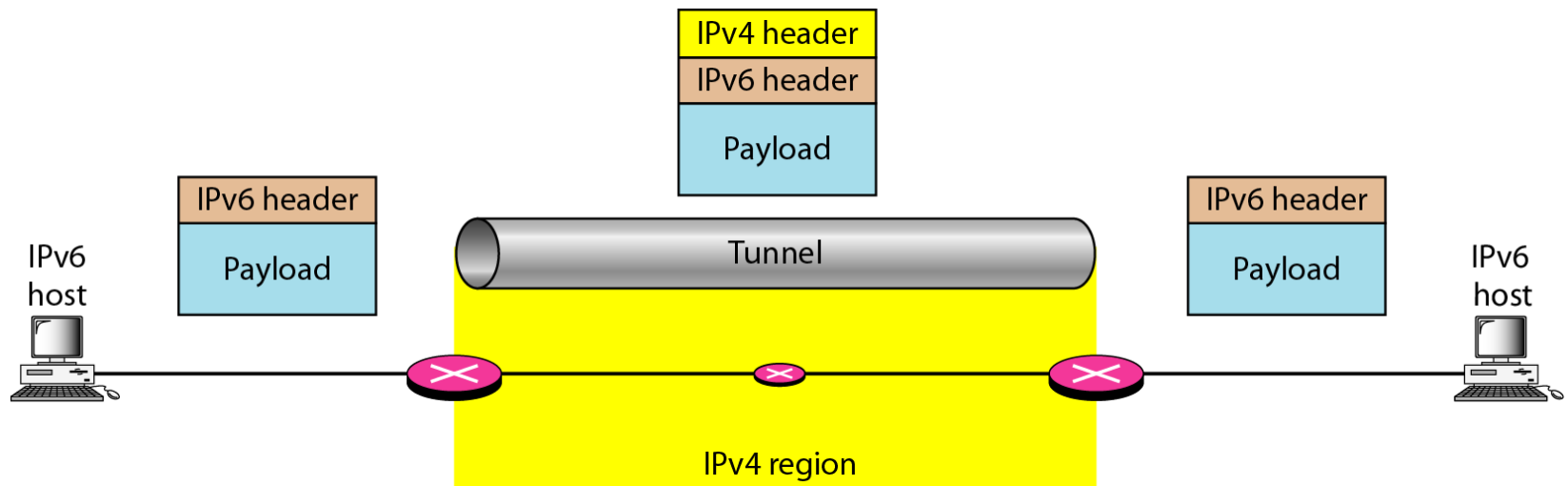
Figure 20.19 *Dual stack*



Tunneling

- Tunneling is a strategy used when two computers using IPv6 want to communicate with each other and the packet must pass through a region that uses IPv4.
- To pass through this region, the packet must have an IPv4 address. So the IPv6 packet is encapsulated in an IPv4 packet when it enters the region, and it leaves its capsule when it exits the region.
- To make it clear that the IPv4 packet is carrying an IPv6 Packet as data, the protocol value is set to 41.

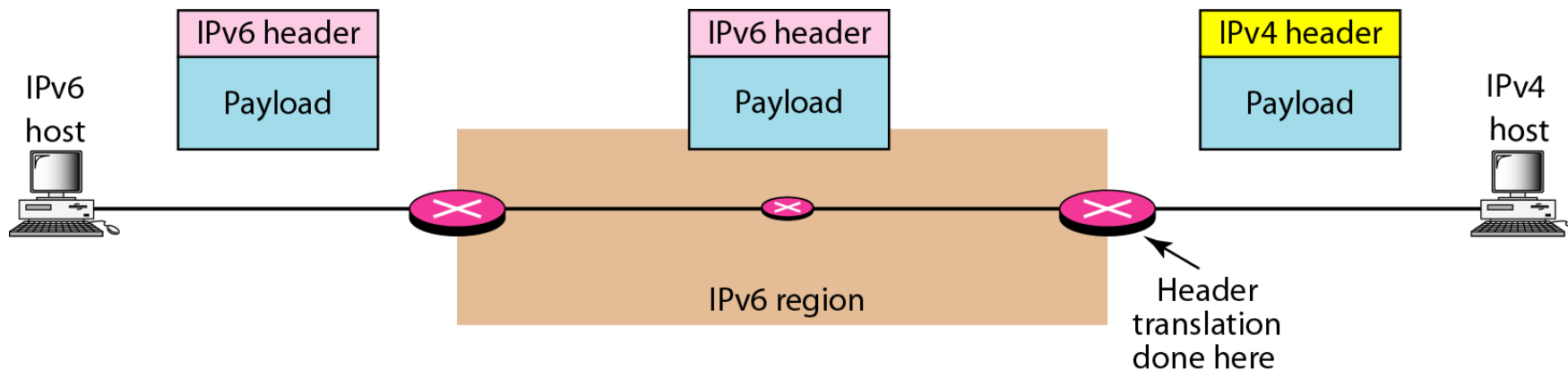
Figure 20.20 *Tunneling strategy*



Header Translation

- Header translation is necessary when the majority of the Internet has moved to IPv6 but some systems still use IPv4.
- The sender wants to use IPv6, but receiver does not understand IPv6.
- In this case, the header format must be totally changed through header translation.
- The header of the IPv6 packet is converted to an IPv4 header.
- Header translation uses the mapped address to translate an IPv6 address to an IPv4 address.

Figure 20.21 *Header translation strategy*



There are some rules used in transforming an IPv6 packet header to an IPv4 packet header.

Table 20.11 *Header translation*

<i>Header Translation Procedure</i>
1. The IPv6 mapped address is changed to an IPv4 address by extracting the rightmost 32 bits.
2. The value of the IPv6 priority field is discarded.
3. The type of service field in IPv4 is set to zero.
4. The checksum for IPv4 is calculated and inserted in the corresponding field.
5. The IPv6 flow label is ignored.
6. Compatible extension headers are converted to options and inserted in the IPv4 header. Some may have to be dropped.
7. The length of IPv4 header is calculated and inserted into the corresponding field.
8. The total length of the IPv4 packet is calculated and inserted in the corresponding field.